

RESEARCH ARTICLE

CoboGesture: A Continuous Hand Gesture Dataset and Recognition Method for Human-Collaborative Robot Interaction

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ABSTRACT The use of collaborative robots (aka cobots) - a type of robot designed to work alongside humans have recently increased. To create a seamless collaboration between human and cobot, hand gesture is the most widely used. In this paper, we focus on addressing the hand gesture recognition for human-cobot interaction. Recently, many models have been proposed for hand gesture recognition using various modalities, including RGB, depth, and skeleton data. However, the majority focus on recognizing isolated hand gestures on a segmented video sequence which limits their practical application. In this paper, we propose two methods for continuous hand gesture recognition, namely RGB-CoGes and SkeRGB-CoGes. The RGB-CoGes method relies solely on RGB image sequences, while SkeRGB-CoGes integrates both skeleton and RGB modalities. Both methods share a common training phase, in which an isolated gesture recognition model based on VideoMAE and VideoMAEv2 is trained on segmented sequences. During inference phase, RGB-CoGes employs a sliding window approach on continuous sequences to classify gestures within each window using the trained model. In contrast, SkeRGB-CoGes leverages the lightweight skeleton information to detect the start and end points of gestures, after which the same trained model is applied to the identified segments. Additionally, to evaluate and promote the research on continuous hand gesture recognition, a self-built dataset named CoboGesture containing 19 different hand gestures in the context of human-cobot interaction has been collected and fully annotated. Experiment results demonstrate the effectiveness of the proposed framework in enhancing gesture recognition performance for human-cobot interaction. Regarding continuous recognition, the proposed methods RGB-CoGes and SkeRGB-CoGes achieved frame-wise accuracies of 0.87 and 0.90, and overlap scores of 0.68 and 0.67, respectively, significantly outperforming the skeleton-based methods. Furthermore, we demonstrated the successful deployment of the proposed method on an edge device.

INDEX TERMS Hand gesture, human-cobot interaction, collaborative robots, continuous recognition.

I. INTRODUCTION

Nowadays, with the rapid development of science and technology, robots have been widely adopted in numerous fields, such as educational, social, therapeutic and amusement

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sectors. More and more types of robot are generated to help humans to do some complicated and difficult tasks. In fact, robots are capable of performing dangerous tasks and can work longer hours without fatigue. This makes them particularly valuable in industries such as manufacturing and construction, where hazardous environments and repetitive tasks pose risks to human workers. In production lines,

industrial robots are typically set up to work independently, enclosed behind fences to ensure human safety. Meanwhile, a collaborative robot (cobot) is designed to work alongside humans through a human-cobot interaction process. Cobot is the representative genre of robot with the fastest development and have been receiving more and more attention of research community in the field of smart robot. Not only are cobots safer to collaborate with, they are also designed to easily adapt from one purpose to another just by replacing the end-effector [1], [2].

In traditional methods, humans usually used additional hardware devices, such as hand-held controllers or computers with keyboard and mouse, to interact with robots. These methods make operators feel uncomfortable because they can not directly work with robots while performing their tasks. Moreover, extra devices and software come with additional costs, and the fact that robots cannot adapt well to multiple projects makes working with them less appealing to small-scale factories. These drawbacks resulted in the development of promising and cost-effective techniques without any additional device, called vision-based technology. Vision-based human-cobot interaction offers numerous benefits that enhance the usability, efficiency, and safety of robotic systems in various applications. This approach allows workers to freely interact with cobots without any equipment and involves the ability of robots to respond to human hand gestures. This task, also known as computer vision for robots, is a crucial process that enables robots to identify and respond to the complex human behavior. Visual information is captured through a camera system mounted on robots, then used to help them understand the activities required by humans and make informed decisions on how to interact or assist the person. In the literature, many studies have aimed to achieve both high accuracy and real-time response in human-cobot interaction tasks [3], [4], [5], [6], [7], [8], [9], [10], [11]. Despite these efforts, several challenges persist when these systems are deployed in real-world environments. These challenges include the diversity of human activities, background clutter, varying lighting conditions, different camera viewpoints, and partial occlusion. To capture visual data for hand gesture recognition (HGR), various types of cameras are employed, such as RGB cameras, thermal cameras, and infrared cameras, each offering unique advantages and limitations based on the application scenario.

It is worth noting that existing HGR methods are generally divided into two categories: isolated recognition and continuous recognition. The isolated recognition approach focuses on dealing with video segments containing only a single hand gesture and treats the task as a video classification problem. The objective is to detect and classify a specific gesture within a pre-defined video segment. In contrast, continuous recognition involves processing multiple gestures in real-time and generating results that correspond to a sequence of gesture labels over time. This task is significantly

more challenging than isolated recognition, as it requires not only classifying but also determining the precise timing and duration of each gesture within a continuous video stream. Accurate temporal localization of gestures is critical for continuous recognition, making it a more complex problem.

This paper introduces a deep-learning framework designed to handle continuous hand gesture for human-cobot interaction task. The proposed framework comprises two crucial stages that are isolated recognition and post-processing. While isolated recognition operates as a video classifier, identifying individual gestures within segmented sequences. Post-processing aims at refining the isolated results to ensure smooth and accurate outputs for continuous gesture recognition, thereby enhancing the system's overall performance in real-time applications. For isolated recognition, transformer-based VideoMAE [12] is exploited and its decoder is replaced by a linear convolutional layer to predict the class for each interested video input. These isolated results are fed forward to post-processing to provide the final results for continuous recognition.

Additionally, the lack of training data leads to a low performance for any computer vision problem. In fact, there are a few publicized hand gesture datasets for continuous recognition task. This is the motivation for us in this work to build a novel continuous hand-gesture dataset for human-cobot interaction. Inspired by the gesture design of Barattini et al. [13], our self-built dataset named CoboGesture consists of total 150 untrimmed videos of 50 subjects with 19 hand-gestures. To resemble an industrial environment, dataset is collected in a laboratory with many computers and electronic devices in the background, and flashing illumination. Cluttered background and bad lighting conditions can pose some challenges to the dataset, requiring the HGR method to be robust with the environment.

In summary, the contributions of this paper are two-fold.

- First, two methods for continuous hand gesture recognition, namely RGB-CoGes and SkeRGB-CoGes are proposed. The RGB-CoGes method relies solely on RGB image sequences, while SkeRGB-CoGes integrates both skeleton and RGB modalities. Both methods share a common training phase, in which an isolated gesture recognition model based on VideoMAE and VideoMAEv2 is trained on segmented sequences. During inference phase, RGB-CoGes employs a sliding window approach on continuous sequences to classify gestures within each window using the trained model. In contrast, SkeRGB-CoGes leverages the lightweight skeleton information to detect the start and end points of gestures, after which the same trained model is applied to the identified segments.
- Second, a new dataset CoboGesture for continuous recognition, containing 19 different hand gestures designed for human-cobot interaction, is collected and

fully annotated. The dataset is made available to the research community at the web link.¹

The remaining of the paper is organized as follows. Sec. II analyzes some works related to HGR. Then, details of our dataset CoboGesture, including list of gestures, statistics and data collection method, are described in Sec. III. The proposed framework is described in details in Sec. IV. Experimental results are presented in Sec. V

II. RELATED WORKS

Some early studies on HGR problem have relied on data collected from hand-held devices integrated into wristbands or hand gloves [14], [15]. However, this type of data suffers from noise and inconvenience, making this approach unappealing in industrial environments. Consequently, recent research has focused more on vision-based approaches, using cameras as the data source. This approach is not only more effective and explainable but also allows for the inheritance of previously studied architectures and techniques in computer vision. In this section, several remarkable studies related to vision-based HGR are briefly discussed. Additionally, several existing datasets for HGR task will be also analyzed.

A. HAND GESTURE RECOGNITION

1) ISOLATED HAND GESTURE RECOGNITION

To recognize the isolated gestures, some early studies leverage the ability of 3D-CNN models for building spatio-temporal representation. In the work of Li et al. [16], a spatio-temporal attention-based ResC3D model was introduced to learn a compact and effective representation for hand gestures by leveraging both spatial and temporal domains. This model combines the strengths of a residual network with a C3D model to enhance recognition performance. Similarly, in [17], the authors proposed a multi-modal approach where a uni-modal 3D CNN was used for feature extraction, supported by multi-modal training knowledge. Additionally, Spatio-temporal Semantic Alignment (SSA) loss was applied to optimize the model's performance.

Although 3D-CNN models have achieved certain results, they are really complex and inefficient due to capturing both spatial and temporal information. Based on this observation, Yu et al. [18] proposed a 2D-CNN modal to fuse multi-modal data, RGB for spatial and optical flow for temporal information. In [5], a Multi-scale Parallel Convolutional Neural Network (MPN) with high-resolution and low-resolution channels was introduced, using RGB, depth, and EMG signals to enhance recognition. In [19], the author introduced a 2D-CNN and LSTM model combination to learn short-term and long-term video features, with RGB and optical flow fused for 2D-CNN input, followed by LSTM for sequence learning. 3D-CNN and ConvLSTM networks have been also proved to be effective for dynamic hand gesture recognition [4], [20].

Existing methods using RGB images often face computation complexity issues, leading to inefficiency. To address this, some work focuses on extracting skeleton data to reduce complexity while maintaining recognition rates [21], [22], [23]. In [24], the authors proposed a CNN-based model estimates 3D hand keypoints, which are then projected into three views for better hand gesture representation. This framework uses 3D-CNNs to learn spatial and temporal information, improving recognition rates. In [22], a combination of OpenPose and ST-GCN was used for isolated hand gesture recognition, with OpenPose capturing skeleton information and ST-GCN classifying gestures. Inspired by ST-GCN, [21] developed a new hand skeleton model using three types of edges to represent joint actions. Despite significant progress, this approach only handles segmented videos, limiting its real-life application. Recently, research has shifted towards continuous hand gesture recognition.

2) CONTINUOUS HAND GESTURE RECOGNITION

Compared to isolated hand gesture recognition, the continuous approach is considered more realistic and challenging. Continuous hand gesture recognition methods must not only classify the activity but also accurately determine its start and end times. These methods typically follow a two-step framework. In the first step, isolated hand gesture recognition is applied within a sliding window to assign an activity label to the observed segment. Then, results are refined by a post-processing step [5], [25], [26], [27], [28]. In the literature, the sliding window is the most widely used techniques for video segmentation. Isolated recognition methods are applied to each video segment to predict the corresponding hand gesture. In [26], the authors introduce a general pipeline for continuous hand gesture recognition. A sliding window is used to split an input video into several segments. Each of these segments will be analyzed to predict whether it corresponds to a hand gesture. This approach allows the model to process continuous video streams and identify gestures as they occur within the segmented frames. Additionally, a prediction will be filtered if its confidence score is smaller than a given threshold. This helps to ensure safety in human-cobot interaction. To enhance the accuracy of continuous hand gesture recognition, a majority voting mechanism is proposed in the work of [27]. This involves saving predictions from a fixed time window into a voting box. For each step, the final prediction is determined by the class with the highest number of votes within that window. This method helps in smoothing out the predictions and reduces the impact of noise, leading to more reliable and consistent gesture recognition results. In [28], the authors have made a great effort to build an effective framework based on skeleton information to address continuous hand gesture recognition. This framework includes two crucial phases: hand detection and hand recognition. With the use of TD-Net (Triple Feature Double Motion) [29], a lightweight and robust recognition model, the obtained recognition rates

¹<https://sigm-see.github.io/datasets/CoboGesture.html>

are comparable to some state-of-the-art results on the IPN dataset [30].

From the above analysis, it is evident that most existing methods have primarily focused on isolated hand gesture recognition, while continuous gesture recognition remains relatively underexplored.

B. CURRENT DATASETS FOR HGR

Hand gesture recognition is a rapidly growing field in recent times. Current hand gesture recognition models have achieved high accuracy and can be applied in various settings such as entertainment, medical, and industrial applications. However, this field still faces significant challenges including the complexity of gestures, variations among individuals, and diverse background environments.

The availability of hand gesture datasets was limited until 2007 and has only increased recently. To date, many datasets have been publicized, providing a large number of frames across multiple modalities (RGB, Depth, Optical Flow, Skeleton, etc.). However, most gestures in these datasets are originally defined for interacting with screens (e.g., swipe, tap) or are related to daily life activities (e.g., sweeping the house, pouring milk) rather than for interaction with robots. In this section, some existing datasets for hand gesture recognition are briefly mentioned.

IPN-Hand [30] is a large-scale benchmark dataset collected from 50 different subjects, consisting over 800,000 frames and 4,000 gesture instances. All these frames are recorded at a frame rate of 30 FPS with a resolution of 640×480 pixels. In this dataset, each participant performs continuously 21 gestures with three random breaks. Among them, 13 gestures are specifically designed for interaction with touchless screen in both static and dynamic scenarios. To date, IPN-Hand is considered as one of the most challenging datasets for gesture recognition because of strong variations in background and lighting conditions. Additionally, even though originally built for touchless screen manipulation, the defined set of gestures can still be mapped to a new set of commands for HCI tasks, despite the potential loss of semantics.

InHARD [31] is an industrial dataset for Human Action Recognition, created to facilitate Human-Robot Collaboration in industrial environments. This dataset involves 16 subjects performing in the assembly task with the help of a UR10 Cobotic arm, each subject conducted the task two to four times. It includes video and skeleton modalities collected from three different points of view. InHARD dataset encompasses 13 main actions combined with 16 common objects in the industrial workplace, resulting in a total of 72 high-level classes that describe the actions more accurately. The InHARD dataset primarily focuses on recognizing workers' actions in an industrial environment, rather than capturing the interaction between the worker and the collaborative robot (cobot). Additionally, to accurately capture the human skeleton, workers are required to wear

motion sensors, which can occlude parts of their hands. As a result, RGB-based methods may face challenges in this context

SHREC [32] is a 3D dynamic hand gesture dataset that provides sequences of hand skeletal data combined with depth images. This dataset contains 14 dynamic hand gestures collected with Intel RealSense camera from 28 subjects. Each gesture is performed from 1 and 10 times by each subjects in two scenarios: using one finger and using the whole hand, resulting in 2,800 image sequences with a resolution of 640×480 . However, this dataset support only the isolated gesture evaluation.

Jester [33] was introduced in 2019 and, at that time, was the largest vision-based hand gesture dataset. It consists of 148,092 short clips, each 3 seconds long, resulting in more than 5 million frames. A total of 1,376 volunteers were involved in data collection, under significant variations in the background. Similar to the SHREC dataset, the Jester dataset is dedicated to evaluating isolated gesture recognition.

NVGesture [34] is designed for touchless driver control applications and recorded with three modalities: RGB, Depth, and Infrared. It includes a total of 1,532 dynamic hand gestures of 25 gesture classes collected from 20 subjects. This dataset focuses on gesture recognition in automotive environments, using RGB-D and infrared to collect data.

HANDS [35] is an RGB-D dataset designed for HCI purposes with 15 static gestures. Depth frames are recorded using a Kinect v2 sensor and later aligned with RGB frames. Despite the dataset's various backgrounds, lighting conditions, and changing subject positions, it has limitations. The dataset is recorded with only 5 subjects, each corresponding to 2400 RGB-D frames, resulting in a total of 400 seconds of clips. Additionally, videos recorded from subjects are filtered to discard blurred and occluded frames and segmented to 150 frames per gesture, which inadvertently disables training and benchmark on continuous HGR.

VIVA challenge's dataset [36], [37] is a multi-modal dynamic hand gesture dataset designed for studying natural human activities in real-world driving environment. This dataset involves 885 intensity and depth video sequences collected by a depth camera Microsoft Kinect. These sequences performed by 8 subjects inside a vehicle and classified into 19 different dynamic hand gesture classes. A key feature of the VIVA dataset is its challenging conditions, including cluttered backgrounds, fluctuating illumination, and frequent occlusion, making it ideal for evaluating the robustness of gesture recognition methods.

EgoGesture [38] is a multi-modal large-scale dataset built for egocentric hand gesture recognition. This dataset has a total of 2,081 RGB-D videos of 50 distinct objects recorded from six complex indoor and outdoor scenes with strong variations in illumination and background. These videos result in 24,161 gesture samples divided into 83 classes of both static and dynamic gestures, and are used for both isolated and continuous hand gesture recognition.

TABLE 1. Description of publicly available datasets for hand gesture recognition and our dataset CoboGesture.

No.	Name	Subjects	Classes	Samples	Modalities	Tasks	Application
1	IPN-Hand [30]	50	13	4,218	RGB	Isolated/Continuous	Touchless screen interaction
2	InHARD [31]	16	13	-	RGB+skeleton	Isolated/Continuous	Industrial actions of workers.
3	SHREC [32]	28	14	-	RGB-D+skeleton	Isolated	Human-computer interaction
4	Jester [33]	1,376	27	148,092	RGB	Isolated	Human-computer interaction
5	NVGesture [34]	20	25	1,532	RGB-D+Infrared	Isolated/Continuous	Automotive interfaces and touchless control systems
6	HANDS [35]	5	15	400	RGB-D	Isolated	Human-computer interaction
7	VIVA [37]	8	19	885	RGB-D	Isolated/Continuous	Driving environments.
8	EgoGesture [38]	50	83	24,161	RGB-D	Isolated/Continuous	Interaction using egocentric vision
9	CoboGesture	50	19	2,479	RGB+Skeleton	Isolated/Continuous	Human-cobot interaction

Table 1 summarizes various existing datasets and the dataset CoboGesture introduced in the paper, providing details such as the number of classes, subjects, samples, modalities, tasks, and their respective applications.

As analyzed, several datasets have been collected for hand gesture recognition. Some of these datasets are defined for touchless screen interaction and driving interaction. Among the aforementioned datasets, InHARD was collected in the context of human-robot collaboration. However, this dataset focuses on industrial actions rather than gestures for interacting with robots. Therefore, we aim to create a new hand gesture dataset specifically for human-cobot interaction. The process of collecting and annotating the data will be described in the next section.

III. CoboGesture - A NEW HGR DATASET FOR HUMAN-COBOT INTERACTION

A. MAIN REQUIREMENTS AND GESTURE SET DESIGN FOR HUMAN-COBOT INTERACTION

In the industrial environment, production lines are dynamic, with equipment and working positions frequently updated to accommodate different projects. This environment is often noisy, with machinery sounds and alarm signals making verbal control of robots challenging. While solutions like attached microphones or wearable sensors, such as inertial measurement units embedded in wristbands or hand gloves, cause inconvenience to the users, especially in the environment having a narrow working space. Furthermore, the scenario in which one operator takes over from another must be considered. In such cases, the new operator will often not have the necessary tools and equipment immediately at hand. Exchanging devices between operators can be time-consuming and may also lead to issues such as size mismatches. It is evident that the use of additional devices presents significant drawbacks in the field of human-cobot interaction.

Thanks to the successes in vision-based approaches in recent years, an image dataset would be preferred over wearable sensors, which are uncomfortable and less intuitive. Compared to vocal commands that can only be recognized

from a short distance, gestural commands are understandable at a wider range, as long as the camera provides an image of good enough quality. Considering these factors, a vision-based dataset for hand gesture recognition in industrial settings must be developed. This dataset would help address the challenges of gesture complexity, variations among individuals, and diverse background environments, ensuring effective human-robot collaboration.

Based on the above analysis, in this paper, we aim at building a continuous hand gesture dataset for human-cobot interaction, namely CoboGesture dataset. With the objective of fostering collaboration between workers and robots, the dataset must ensure to meet the following criteria:

- **One-Handed Gestures:** Workers are expected to interact with robots while holding tools. Therefore, gestures should be performed with one hand, allowing for either right or left hand usage to accommodate different preferences.
- **Versatility Across Various Robots:** The dataset should be applicable to a broad range of robotic applications. Consequently, the proposed gesture set must be comprehensive enough to cover the use cases for these two types of robots.
- **Ease of Learning:** The gesture set should be compact and straightforward to learn, minimizing the training time and effort required for new operators. The gestures can be either static or dynamic but must be intuitive and contextually appropriate. For example, pointing fingers to indicate direction or nodding to signify agreement.

By adhering to these requirements, the dataset will be practical and effective for enhancing human-robot collaboration in industrial environments. This approach ensures that the dataset addresses the unique challenges of the industrial workspace while being intuitive and easy to implement for operators.

In [13], a set of 19 hand gestures have been carefully designed by taking into account all aforementioned requirements. Therefore, in our study, we collect a video dataset of these 19 gestures. Additionally, an extra class 'No-gesture' is



FIGURE 1. Scene for data collection.

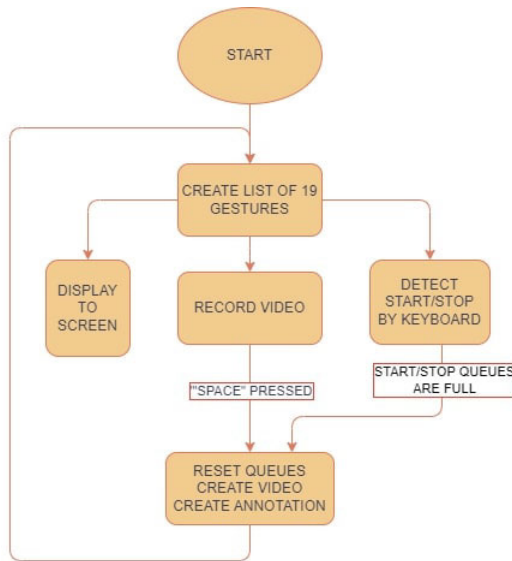


FIGURE 2. Flowchart of the data collection scheme.

also considered as a gesture having either no action or action differing from 19 those gestures.

B. DATA COLLECTION

To collect data, an industrial FLIR camera is used to record RGB videos with a resolution of 1440×1080 at the rate of 30 FPS (frame per second). The camera is positioned 2.5 meters away from the subjects, with its height adjusted between 1.1 to 1.2 meters, ensuring both hands are always visible in the video. The background is cluttered with various devices such as machines, computers, and tables, closely simulating an actual industrial environment. Figure 1 illustrates the setup used for collecting data for the CoboGesture dataset and Figure 3 shows some examples collected in this environment.

Additionally, to efficiently record and annotate videos, we designed a recording procedure and a script, allowing the

TABLE 2. Example of an annotation file.

ID	Label	Starting frame	Stopping frame
16	Joystick	74	122
17	Identification	199	235
5	FollowMe	310	403
1	Stop	474	564
14	Again	678	731
18	Change	783	815
9	Look	996	1051
11	DepositPart	1167	1222
4	Done	1274	1370

annotation process to be performed simultaneously with the video recording. The flowchart of the software can be seen in Fig. 2. The data collection is conducted in a laboratory equipped with a computer, a camera, and a screen. Additionally, a person, referred to as the annotator is responsible for marking the start and stop times of hand gestures by pressing buttons on a keyboard. To ensure objectivity in the data collection process, the 19 hand gestures are randomly shuffled for each subject. Pre-recorded clips corresponding to these hand gestures are displayed on the screen, allowing the subject to easily follow along. Simultaneously, the annotator presses the “A” and “B” buttons on the keyboard to define the start and end times for each hand gesture. After a participant finishes performing all 19 gestures, the labeling person presses the “Space” button to create a corresponding video and annotation. Each subject is required to perform the gestures in the three different sessions, resulting in three corresponding uAfter data collection, the annotation files are manually reviewed to correct inaccurate annotation information. Table 2 describes an example of an annotation file whereas Fig. 3 illustrates some examples of gesture sequences.

C. CHARACTERISTICS OF CoboGesture DATASET

The dataset consists of 150 videos from 50 subjects, including 11 females and 39 males. Each video ranges from 100 seconds to 140 seconds in length and contains 2,479 gesture samples divided into 19 classes. The time interval between two gestures is approximately 4 seconds. In total, these 150 videos result in 490,000 RGB frames, corresponding to about 4.5 hours of video. Figure 4 shows statistical information of CoboGesture dataset in term of the number of samples and the average duration of each gesture class. From the chart on the left side of Figure 4, we can observe the balance between the 19 gesture classes, which provides advantages for both the training and testing phases. In most gesture classes, the average number of samples corresponding to each gesture is about 150-160. Exceptions are the ‘Follow Me’ and ‘Report’ gestures, which have the smallest average number of samples at 120 and 138, respectively. The ‘Start’ gesture has the largest average number of samples at 188. The right chart of Figure 4 provides information about the average duration measured in frames for each gesture class. It can be seen that each gesture is performed from 45 to 100 frames.

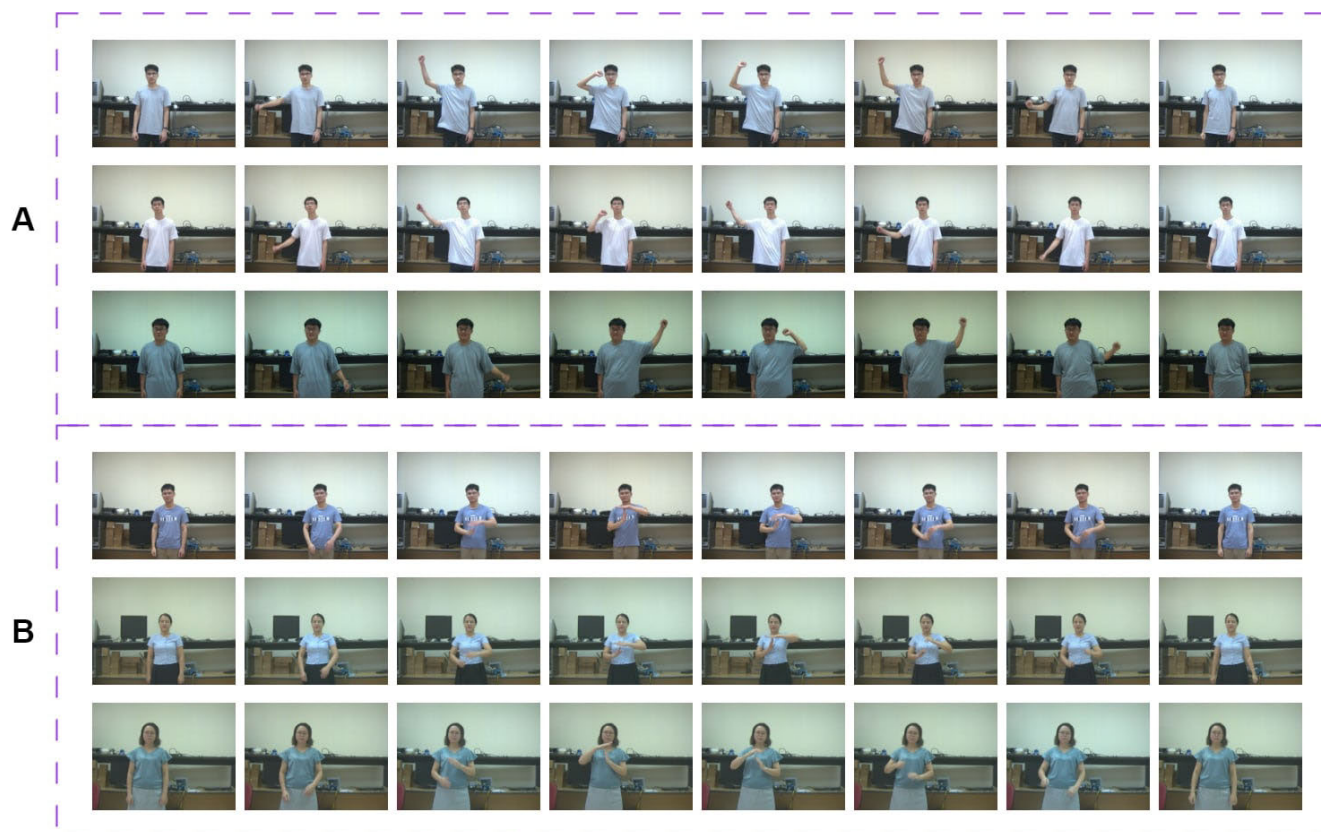


FIGURE 3. Some sequence examples in CoboGesture dataset. (A) 'Start' gesture; (B) 'Joystick' gesture.

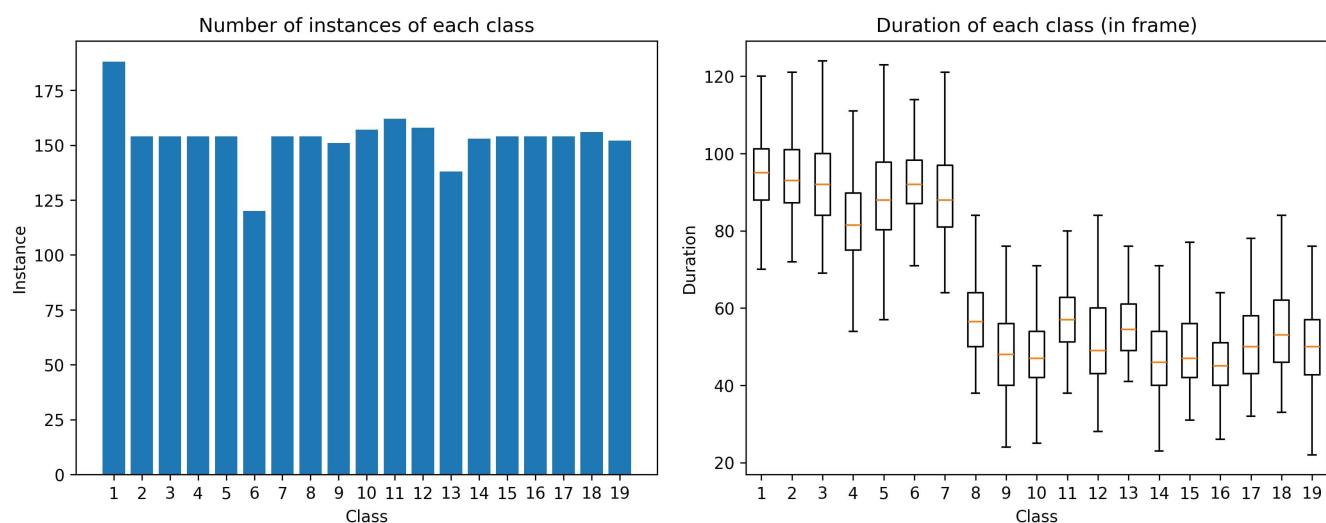


FIGURE 4. Statistics of CoboGesture dataset.

In comparison with other existing hand gesture datasets, CoboGesture has some significant differences and more suitable for industrial environment. Firstly, most gesture datasets are designed for touchless screen interaction or automotive applications, focusing primarily on hand and finger movements. In contrast, the gestures in CoboGesture

dataset involve the movements of the arm, hand, and fingers. This broader range of motion allows these gestures to be observed more clearly and enhances recognition performance. Secondly, in some current studies, the authors declare that a set of gestures which are originally built for touchless screen can be adapted for use in human-cobot

interaction. However, these mappings are often unnatural and lack practical relevance, causing some difficulties for the workers themselves to perform and interact with Cobot. Meanwhile, CoboGesture dataset is specifically designed to reflect tasks commonly associated with collaborative robots in a production line, such as ‘PickPart’, ‘DepositPart’, ‘Slower’, ‘Faster’, etc. These gestures are considered more intuitive and practical for real-world applications. Among the gesture datasets listed in Table 1, the InHARD dataset is specifically designed for industrial environments. However, it primarily focuses on recognizing workers’ actions rather than interaction gestures with collaborative robots. InHARD includes data collected from 16 distinct subjects, whereas our dataset covers 19 gesture classes performed with 50 different subjects, offering a broader and more diverse range of interactions. Especially, some gestures have high similarity in term of both motion and hand shapes, such as ‘Report’ and ‘Pickpart’. This also poses some challenges for both isolated and continuous recognition tasks. Lastly, CoboGesture is recorded in a laboratory equipped with machinery to resemble a real industrial environment, which includes some factors like complex background, lighting variations, etc.

The CoboGesture dataset serves as a benchmark datasets for evaluating isolated and continuous gesture recognition. The dataset is made available for research community. For CoboGesture, we define an evaluation protocol where video sequences of odd ID subjects are used for training while those of even ID subjects are used for testing.

IV. PROPOSED METHOD FOR CONTINUOUS HAND GESTURE RECOGNITION

In this paper, we propose two methods for continuous hand gesture recognition methods named RGB-CoGes and SkeRGB-CoGes. RGB-CoGes relies only on RGB image sequences whereas SkeRGB-CoGes leverages both skeleton and RGB images. Figure 5 illustrates RGB-CoGes and SkeRGB-CoGes methods. The two methods share the training phase where an isolated recognition model is trained on segmented sequences. In the inference phase, RGB-CoGes applies the sliding window technique on the continuous sequences and determine the gesture class within the windows based on the trained model. Finally, several post-processing techniques are employed to refine the results and generate the final continuous gesture recognition output. In SkeRGB-CoGes, to leverage the lightweight of skeleton, we employ skeleton information to order to detect the starting and ending point of a gesture. Then, the trained model is also applied in the determined sequence. Finally, some voting techniques are applied to refine the recognition results.

A. ISOLATED GESTURE RECOGNITION

VideoMAE model [12] is a deep learning model designed for self-supervised learning on video data. It is inspired by the success of MAE (Masked Autoencoders) [39] used for image-based tasks but extend this concept to video

processing. The main objective of VideoMAE is to efficiently learn spatio-temporal representations from unlabeled video data by masking a large portion of input frames and training the model to reconstruct the missing content. This process of masking and reconstructing forces the model to gain a deeper understanding of the structure and semantics of the video, enhancing its ability to generalize and perform various downstream tasks like classification, detection, or segmentation. For this proposed framework, VideoMAE model is treated as a video classifier for isolated recognition.

The structure of VideoMAE model proposed in [12] consists of two crucial component that are encoder and decoder. In comparison with MAE model, VideoMAE has to address with two more challenging issues: (1) temporal redundancy and (2) temporal correlation. It is clear that there is no significant difference between consecutive frames in a given input video and result in redundant information. To overcome this challenge, a temporal down-sampling block is adopted to reduce the number of input frame as well as enhance the efficient video pre-training. This helps save computational resources while still effectively capturing meaningful information from the frame sequence. As described in Figure 5, given an input video includes τ frames, then T frames are chosen through sampling process with a pre-defined temporal stride parameter, referred as stride (in this work, stride = 4), resulting a set of images $I \in \mathbb{R}^{C \times T \times H \times W}$. After that, these sampled frames are divided into some non-overlapping patches and masked with high ratio at 95% before forwarding to the encoder of VideoMAE model. Noted that, for VideoMAE tube masking is applied over frames in the input sequence. The remaining blocks are then flattened via a 3D convolutional layer with a stride size of $(2 \times 16 \times 16)$, where each block has the same size as the stride and is referred to as a token embedding. Additionally, each token is combined with its positional information before being passed to the encoder stage.

For the encoder block, ViT-L/16 [40] is utilized as backbone to learn the representation embeddings from the unmasked input data. ViT-L/16 is a “large” version of Vision Transformer [40], pre-trained on the Kinetics-700 dataset [41]. It comprises 24 layers and operates with a patch size of 16×16 , converting input data into patch embeddings that capture the video’s conceptual information. To handle the temporal correlations between consecutive frames, a space-time attention block is integrated into the model. This block allows the model to learn both spatial and temporal information simultaneously, which results in more effective cube embeddings for the input video. Moreover, the Space-Time Attention block plays a key role in understanding the dynamic nature of videos by leveraging information across both dimensions. It’s important to note that while Multi-head Self Attention (MHSA) within the model can be computationally expensive due to the complexity of the attention mechanism, using a masking ratio of 95% significantly reduces the computation burden. By masking a large portion of the input data, the model focuses on

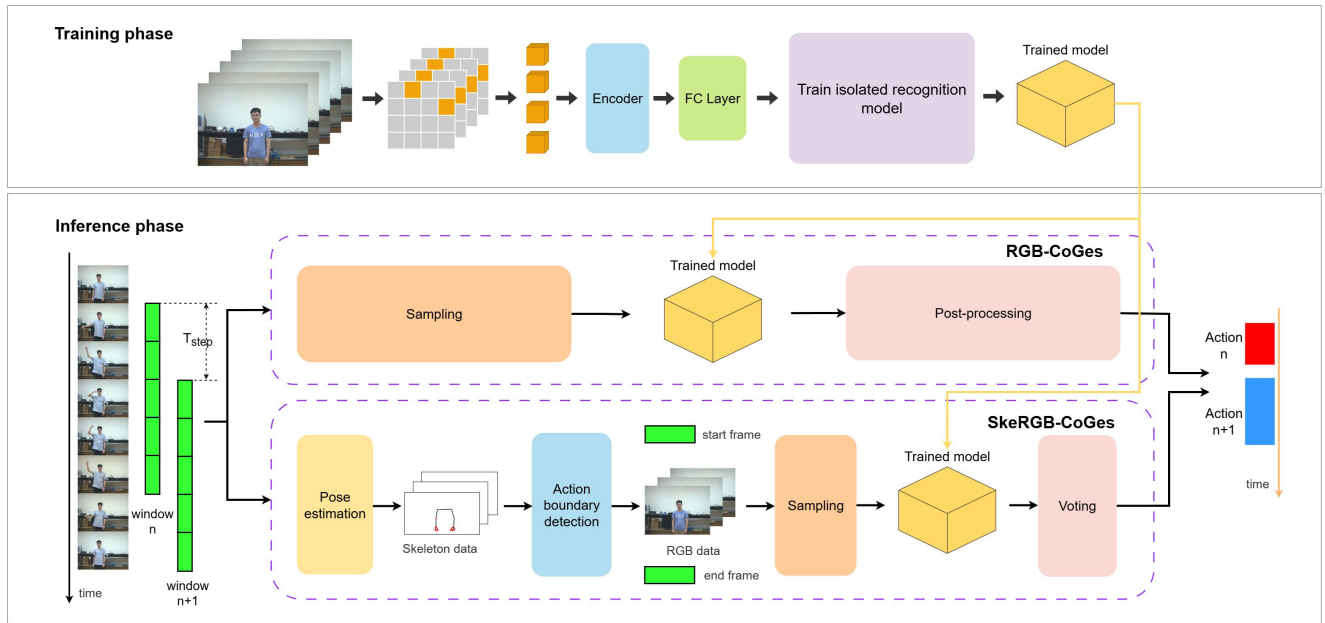


FIGURE 5. Our proposed framework for continuous hand gesture recognition.

learning from limited but crucial visual cues, optimizing both efficiency and effectiveness in the learning process.

The target of the decoder in the VideoMAE model is to reconstruct the original video from the encoded unmasked patches. Typically, this decoder functions as a smaller transformer that processes both masked and unmasked patches, allowing the model to learn meaningful representations by predicting the masked parts of the video. However, in our framework, we modify this model by replacing the decoder block with a linear convolutional layer specifically designed for classification tasks. For this, the linear convolutional layer takes the encoded features and generates output data that are passed through a softmax function. The softmax function converts these outputs into probability distributions across the different classes. The class with the highest probability is then selected as the predicted result, making this modification tailored for classification rather than video reconstruction. This approach simplifies the architecture while focusing on effective video classification. Subsequently, the authors in [42] proposed VideoMAEv2, which incorporates a dual masking strategy. Similar to VideoMAE, its encoder also operates on a subset of video tokens under the encoder mask. However, VideoMAEv2 differs in the design of its decoder. Instead of processing all tokens, it employs a decoder mask that restricts reconstruction to a selected subset of visible tokens. Masking of the decoder reduces the input length and is shown to further decrease computational cost and improve memory efficiency. With these advantages, we also deploy this dual masking strategy of VideoMAEv2, pretrain it with the ViT-L/16 backbone on the Kinetics-700 dataset for a fair comparison. After the self-supervised phase, the model is used as a video classifier and is trained

for isolated recognition, following the training procedure described above.

B. RGB-CoGes: A RGB BASED CONTINUOUS GESTURE RECOGNITION METHOD

In general, continuous recognition builds upon the results of isolated recognition. Consequently, a continuous recognition framework comprises two crucial steps: isolated recognition and post-processing. This combination of isolated recognition and post-processing forms the core of continuous gesture recognition, effectively handling multiple gestures in real-time scenarios. As aforementioned, for isolated recognition, VideoMAE model is used as a classifier that processes segmented videos, treating each gesture independently. In our framework, a sliding window mechanism is exploited to pick up a sequence of frames from the input video. These video segments are considered as the input for isolated recognition. After that, several post-processing strategies aim to refine isolated results for continuous approach. These strategies will be mentioned in more details as follows.

1) SAMPLING

A sliding window continuously slides over the input video stream with a constant step, extracting segments of τ consecutive frames at each step. The size of τ is a crucial parameter, as it determines the temporal span of the gestures being processed. The step parameter is defined with consideration that each action spans multiple frames, but its start and end times may not align exactly with the window boundaries. If the stride is too large, an action may fall between two windows and be missed. Therefore, the step

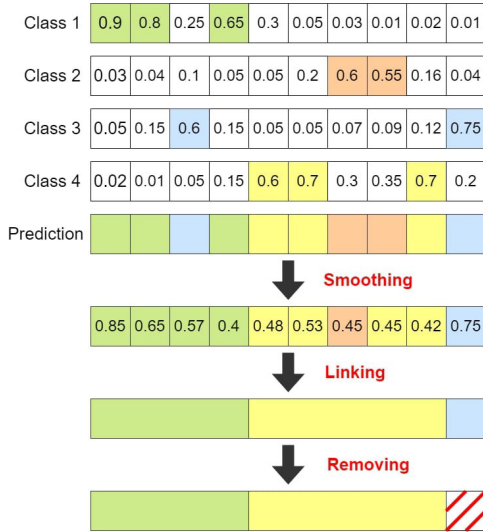


FIGURE 6. Proposed method for post-processing.

is chosen such that the overlap between two consecutive windows is at least 50%. After that, each video segment is sampled to form a sequence of T frames ($T < \tau$) in the same way of temporal sampling strategy for isolated recognition task. Noted that, padding can be added into the end of video through multiplying the last frame.

2) POST-PROCESSING

The post-processing step is essential for refining the results and providing the final continuous recognition output. In this framework, post-processing step involves three main techniques that are smoothing, linking and removing, shown in Figure 6. Given an untrimmed video, proposed model creates a sequence of action probability scores for each frame. Firstly, smoothing strategy aims at filtering out false prediction based on the temporal coherence across consecutive frames. The prediction probability of the current frame is the average probability of it, the previous frame, and the following frame. It excepts for the final frame which the result of probability is equal to the highest prediction probability of frame. Because we add padding by multiplying the final frame into the end of video. Secondly, several discrete segments should be linked to build a complete hand gesture. For this task, a suitable threshold is defined to connect these video segments, called the average time t_{avg} . This means the condition to link two video segments is to have the same label and the gap between the end of the first segment and the start of the second one must not exceed t_{avg} . The value of t_{avg} is selected based on the validation set and represents the average motion duration across all classes in the dataset.

C. SkeRGB-CoGes: A CONTINUOUS GESTURE RECOGNITION COMBINING SKELETON AND RGB SEQUENCES

Taking into account the computation complexity of using only RGB images in real-time scenario, we combine the use of skeleton data with the trained VideoMAE model to

form the SkeRGB-CoGes framework as depicted in Figure 5. This framework is designed for online deployment, where users perform actions directly and the model should return results instantly within a time constraint. Our idea is to use skeleton data first to determine the action boundary, or the start and end points of one gesture in a continuous video stream. The VideoMAE model will be called within that boundary, leveraging RGB data for prediction. After that, a simple voting mechanism is used to select the final label for that action segment. The details of the implementation will be described below.

1) POSE ESTIMATION

In this step, we utilize Mediapipe for the extraction of keypoint coordinates from RGB frames. For each frame, the skeleton data is represented as a tensor with shape $(J \times D)$, which J is the number of keypoints and D denotes the dimensions. For the CoboGesture dataset, we extract 3D keypoints, which include 42 hand joints (21 per hand) and 6 upper-body joints for both left and right wrists, elbows and shoulders. After concatenating, the result tensor for one frame has the shape of (48×3) .

2) ACTION BOUNDARY DETECTION

Building upon a previous work [43], a state-based segmentation module is used to detect the boundary of a gesture by identifying significant changes in hand movement over time. Let $y_j^{(t)}$ and $x_j^{(t)}$ be the vertical and horizontal coordinates of joint j at frame t , and let J_{hand} denote the set of hand joints. At time n , we use a sliding window of τ frames $[n - \tau + 1, n]$, focusing on the first three frames $\tau_{first} = [n - \tau + 1, n - \tau + 3]$ and the last three frames $\tau_{last} = [n - 2, n]$. The state-change conditions are then defined in image coordinates. Given three states: Idle, Moving, Triggered, if the state is Idle and the hand keypoints move significantly upward within the observation window, the state changes from Idle to Moving, marking the start of a gesture.

$$\begin{cases} \max_{t \in \tau_{first}} (y_{wrist}^{(t)}) > y_B \\ \max_{t \in \tau_{last}} (y_{wrist}^{(t)}) < y_B \end{cases} \quad (1)$$

On the contrary, when the hand keypoints depict a downward motion to idle position within the window, the state changes from Moving to Triggered, marking the end of that gesture.

$$\begin{cases} \max_{t \in \tau_{first}} (y_{wrist}^{(t)}) < y_B \\ \max_{t \in \tau_{last}} (y_{wrist}^{(t)}) > y_B \\ \max_{t \in \tau_{last}} (y_{hand}^{(t)}) > y_B \cdot (1 + \text{margin}) \\ \left| \text{mean}_{t \in \tau_{last}} (x_{wrist}^{(t)}) - \text{mean}_{t \in \tau_{last}} (x_{elbow}^{(t)}) \right| < 0.03 \end{cases} \quad (2)$$

where y_B is a threshold defined during the first 3 seconds of calibration before the subject performs any action:

$y_B = y_{\text{wrist}} \cdot (1 - \text{margin})$, and margin is a small offset added to avoid mistaking minor hand movements for actions. Once the module reaches state Triggered, it is then reset to Idle to continue with the next gesture. For a detailed explanation, we refer to the original paper.

3) PREDICTION AND VOTING

After the start point is detected, the trained VideoMAE model is called to process RGB frames, until the end point is identified. The model will process τ frames within the sliding window with a constant step, and those τ frames are also sampled in the same way of temporal sampling as in isolated task before passing to the model. To provide the final recognition output, a simple voting mechanism is deployed instead of post-processing, where the label with the highest frequency of appearance within a detected segment is selected as its final label.

Combining action boundary detection module with this simple voting is expected to enable real-time response of the framework. Moreover, the VideoMAE model is called only within a detected action segment, which could reduce the computational cost for insignificant gesture parts. The feasibility of this framework will be evaluated on the test set and deployment on the edge device, and will be discussed thoroughly in the next section.

V. EXPERIMENTS AND RESULTS

Different experiments are conducted to evaluate the performance of the proposed method at both isolated and continuous gesture recognition steps on our new CoboGesture dataset. We also show a comparison of experimental results with the IPN-Hand dataset [30] and EgoGesture dataset [38]. Additionally, as there are numerous methods have been proposed for skeleton-based human activity recognition, we also perform experiments to compare the RGB-based human gesture recognition with skeleton-based human gesture recognition on our CoboGesture dataset. To achieve this purpose, four methods that are DD-Net [44], TDDNet (Text-Enhanced DD-Net) [45], ST-GCN [46] and CTR-GCN [47] are used. DD-Net model introduced in [44] for isolated recognition has been extended for continuous action recognition [48]. TDDNet is an improved version of DD-Net by integrating a text encoder based on CLIP. ST-GCN is a well-known model capable of learning both spatial and temporal patterns by representing the human skeleton as a dynamic graph. CTR-GCN, on the other hand, learns channel-wise topologies in a refinement manner which demonstrate superiority over other state-of-the-art methods for skeleton-based action recognition. An off the shelf method for human pose estimation that is Mediapipe is applied to estimate the skeleton sequence from RGB sequence.

With our CoboGesture dataset, we utilize NVIDIA Geforce RTX 3090 GPU to train all the models and NVIDIA GeForce GTX 1080 Ti for comparing the model performance on IPN-Hand dataset [30]. All experiments are implemented in Pytorch library [49]. The number of epochs is set to

25, with 5 warm-up epochs and an initial learning rate of 1.25×10^{-4} . For all three datasets IPN-Hand, EgoGesture, and CoboGesture, the values of t_{avg} , τ , step, T are set to 2, 64, 30, 16 respectively.

A. EVALUATION METRIC

In general, hand gesture recognition task consists of two crucial steps: isolated and continuous gesture recognition, different metrics are used to assess the performance of the proposed framework. For isolated recognition which is treated as classification task, Accuracy is employed as an evaluation metric. The formulation of Accuracy is presented as in Equation (3):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (3)$$

where TP, TN, FP and FN are the true positive, true negative, false positive and false negative respectively.

As aforementioned, while isolated recognition is conducted at video segment level, the continuous task is done at the video level. The input for this task is a video that may consisting of multiple gesture instances. For evaluating the continuous recognition, we use two different metrics that are Frame-wise accuracy and temporal action localization.

Frame-wise accuracy refers to the average classification accuracy computed over all frames in a video. Specifically, given a video consisting T frames, let the ground-truth labels of these frames denoted by $G = \{G_1, G_2, \dots, G_T\}$ and their corresponding predicted labels by $P = \{P_1, P_2, \dots, P_T\}$. The frame-wise accuracy is calculated as in Eq. (4):

$$\text{Acc} = \frac{1}{T} \sum_{t=1}^T s(P_t, G_t), \quad (4)$$

where $s(P_t, G_t)$ is defined as follows:

$$s(P_t, G_t) = \begin{cases} 1 & \text{if } P_t = G_t \\ 0 & \text{if } P_t \neq G_t \end{cases} \quad (5)$$

If the set consists of N videos, then Frame-wise Accuracy is computed by the following equation:

$$\text{Frame-wise Accuracy} = \frac{1}{N} \sum_{i=1}^N \text{Acc}_i \quad (6)$$

where Acc_i is the frame-wise accuracy of the i^{th} video that is computed by Equation 4.

Since Frame-wise Accuracy does not account for the temporal information of recognized gesture sequences, which is a crucial aspect in continuous action recognition tasks, we adopt a metric introduced in [50] for continuous driver activity recognition in the AI City Challenge. With a ground-truth gesture sequence g . The start time and the end time of g are denoted as g_s and g_e . The closest predicted gesture sequence p which has the same class as g and the highest overlap score os between p and g is chosen as matched predicted gesture. The overlap os between the

ground-truth gesture sequence g and the predicted sequence p is determined by the ratio between the temporal intersection and the temporal union of the two gesture sequences, which is given by:

$$\text{os}(p, g) = \frac{\max(\min(g_e, p_e) - \max(g_s, p_s), 0)}{\max(g_e, p_e) - \min(g_s, p_s)} \quad (7)$$

As predicted gesture sequence may be shorter or longer than the ground-truth sequence, and its start and end times could differ from the ground truth, to account for these aspects, the authors in [50] added a condition that the start time p_s and end time p_e of the action p can be within the range $[g_s - \theta, g_s + \theta]$ and $[g_e - \theta, g_e + \theta]$, where θ is a predefined threshold. In our study, the value of θ is set to 2s.

After matching each ground truth gesture with at most one predicted activity and processing them in the order of their start times, the overlap score of all unmatched ground-truth gestures and unmatched predicted gestures are assigned to 0. The final score denoted *Score* is computed as an average overlap score across all matched and unmatched gestures. A higher final score indicates a better estimation of the start and end times of the gestures by the model.

B. EXPERIMENTAL RESULTS

1) ISOLATED GESTURE RECOGNITION RESULTS

Table 3 presents the results for the isolated gesture recognition task in CoboGesture. As shown, the proposed method significantly outperforms all skeleton-based models, achieving a significant improvement in accuracy. Specifically, our approach reaches an accuracy of 95.4% for VideoMAE and 96.7% for VideoMAEv2, compared to 84.2% for DDNet, 85.5% for TDDNet, 88.15% for ST-GCN and 92.31% for CTR-GCN. These results clearly demonstrate the effectiveness of the proposed method in recognizing isolated gestures.

In addition to accuracy, Table 3 also reports the number of model parameters and the average inference time per gesture sequence. While the proposed method achieves superior recognition performance compared to skeleton-based approaches, it comes at the cost of increased model complexity. Specifically, it requires 0.123 seconds per gesture sequence, whereas skeleton-based methods achieve significantly faster inference at just approximately 0.044 - 0.059 seconds on the same computing platform. These results highlight the need for further optimization to reduce computational time, making the model more suitable for real-time applications.

Additionally, to provide a more detailed analysis of the methods' performance, the confusion matrices of these methods are illustrated in Figures 7, 8. We can observe that certain gesture classes, such as 'Stop', 'Home', 'Joystick', 'Faster', 'Done', and 'Help', are classified with high accuracy by both DDNet and the proposed method, as they are quite distinct from the others. However, DDNet struggles to recognize gestures like 'Look', 'PickPart', 'DepositPart', and 'Report' due to the high similarity in hand and

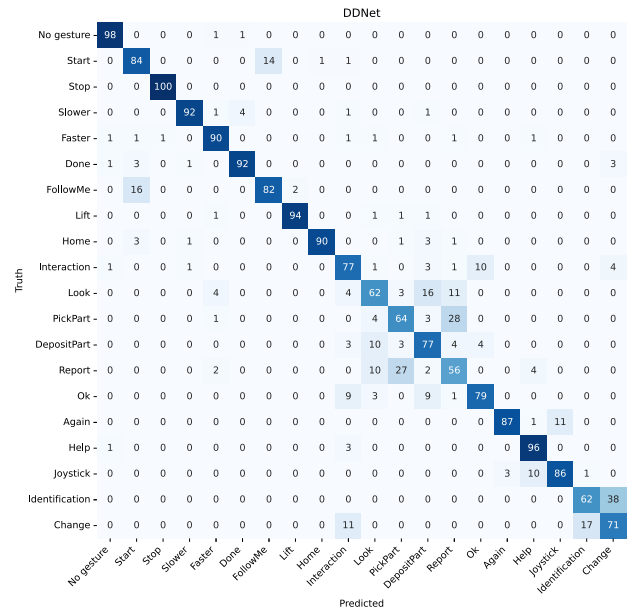


FIGURE 7. The confusion matrix of DDNet for gestures in CoboGesture dataset.

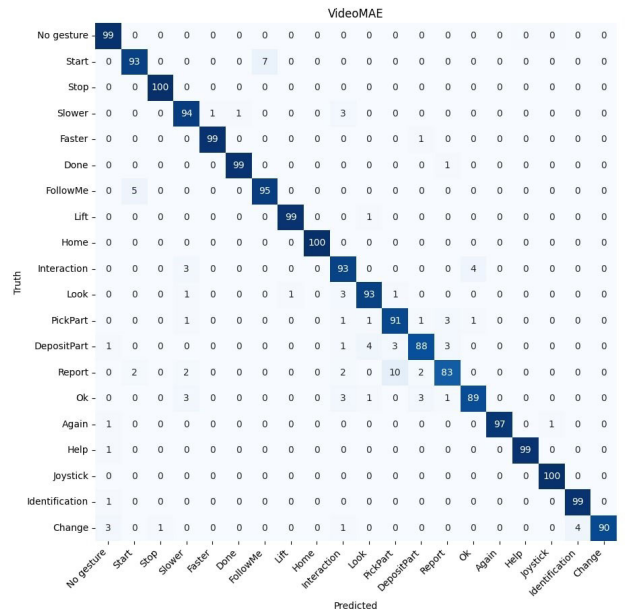


FIGURE 8. The confusion matrix of the proposed method for gestures in CoboGesture dataset.

arm shapes and motion trajectories. Thanks to the rich information obtained from the RGB encoder, the proposed method successfully distinguishes these gestures. However, the results for 'PickPart' and 'Report' remain low, with a 10% misclassifications rate. As illustrated in Figure 9, the most distinguishing parts of these gestures occur in small image regions involving the fingers, which leads to model confusion.

TABLE 3. Isolated recognition results of the state-of-the-art methods and those of the proposed method in CoboGesture dataset.

Models	Input	Accuracy (%)	No.Parameters	Inference time (s)
DDNet [45]	Skeleton	84.2	2.4M	0.057
TDDNet [46]	Skeleton + Text	85.5	2.4M	0.057
ST-GCN [47]	Skeleton	88.15	2.8M	0.059
CTR-GCN [48]	Skeleton	92.31	1.5M	0.044
Proposed method with VideoMAE	RGB	95.4	303.8M	0.123
Proposed method with VideoMAEv2	RGB	96.7	303.8M	0.123

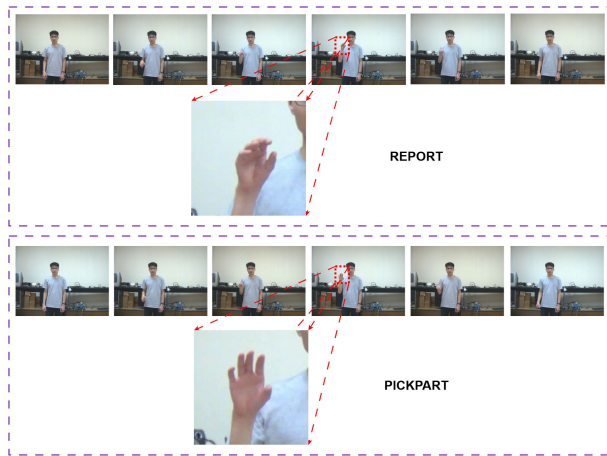
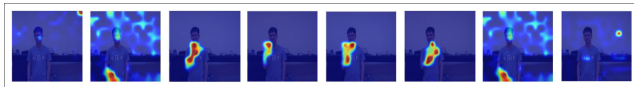
**FIGURE 9.** Examples sequences of two gestures that are 'Report' and 'PickPart' in CoboGesture dataset.**FIGURE 10.** GradCam for 'Stop' sequence.

Figure 10 shows the Grad-Cam visualization [51] for the 'Stop' gesture of a subject in CoboGesture dataset. The highlighted areas represent regions of attention, with darker areas indicating greater focus. In the initial frames, there is little hand movement, so the model does not focus on the regions containing the hands. In the middle of motion, the model focuses on the right hand area to accurately classify the gesture.

Table 4 presents a comparison between the models used for isolated gesture recognition in our proposed methods and several deep learning models on the IPN-Hand dataset [30] and EgoGesture dataset [38].

Across the two datasets, VideoMAEv2 achieved the highest accuracies—84.59% on the IPN-Hand dataset and 89.85% on the EgoGesture dataset—outperforming all other models. Experimental results show that traditional hand-crafted and 2D CNN-based methods, such as ResNet-50, iDT-FV, and VGG16, yield relatively low accuracies, while 3D convolution-based methods like C3D further improve performance by jointly learning spatial and temporal features. Although VideoMAEv2 attains the best results on these

datasets, its accuracy remains lower than that achieved on CoboGesture. This performance gap can be attributed to the higher variability in the IPN-Hand dataset, which includes diverse backgrounds and lighting conditions, making accurate recognition more challenging. Additionally, the EgoGesture dataset is captured from a first-person viewpoint with severe occlusions.

2) ABLATION STUDY

We first conduct an ablation study on the impact of models used for isolated gesture recognition and post-processing steps in continuous gesture recognition. The obtained results are shown in Table 5.

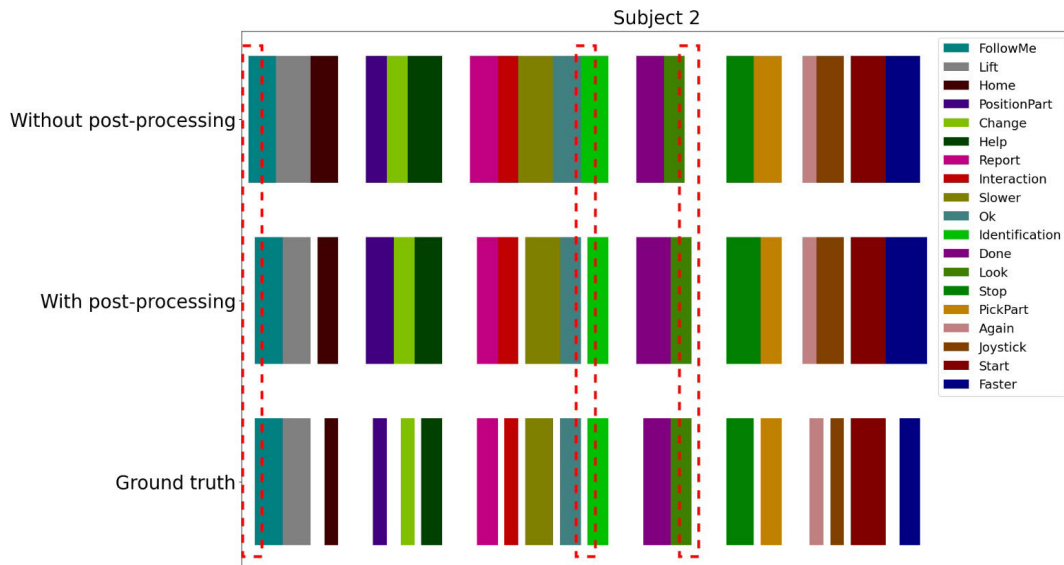
Table 5 shows the performance of our RGB-CoGes method when using VideoMAE and VideoMAEv2 as the backbone for isolated hand gesture recognition. We evaluate both models with and without post-processing. Between the two models, VideoMAEv2 achieves the best frame-wise accuracy of 0.872 and a score of 0.678, slightly outperforming VideoMAE. These results demonstrate that the newer VideoMAEv2 architecture provides a consistent improvement over the original VideoMAE.

Furthermore, we can observe the role of post-processing in continuous gesture recognition performance. Post-processing improves the frame-wise accuracy of VideoMAE from 0.772 to 0.870 and the score from 0.423 to 0.681. For VideoMAEv2, the gain is smaller because the model already produces smoother temporal predictions, but accuracy still increases slightly (from 0.871 to 0.872) and the score improves (from 0.677 to 0.678). This can be explained by the fact that the post-processing step helps eliminate small misclassified segments while merging correctly classified segments into a complete action, leading to more accurate identification of the start and end timestamps of each gesture.

To better demonstrate the model's recognition capability and the improvement in results with post-processing, we visualize a gesture recognition result of a subject in our dataset, as illustrated in Figure 11. We compare the start and end timestamps of each predicted action with the ground truth. In each figure, the top row shows the predicted recognition results, while the bottom row represents the ground truth. We observe that the model classifies actions well, and the post-processing step improves both Frame-wise Accuracy and Score (activity overlap score). However, the start and end points of some gestures are incorrectly identified. This is

TABLE 4. Comparison between the proposed method and other methods for isolated recognition task in IPN-Hand dataset [30] and EgoGesture dataset [38].

Dataset	Model	Accuracy (%)	No. Parameters
IPN-Hand [30]	ResNet-50 [53]	73.10	46.25M
	C3D [54]	77.75	50.75M
	Proposed method with VideoMAE	78.81	303.8M
	Proposed method with VideoMAEv2	84.59	303.8M
EgoGesture [38]	iDT-FV [55]	64.30	-
	VGG16 [56]	62.50	-
	VGG16+LSTM [57]	74.70	-
	C3D [54]	86.40	-
	Proposed method with VideoMAEv2	89.85	303.8M

**FIGURE 11.** Visualization of continuous gesture recognition results when using VideoMAE model without and with post-processing.**TABLE 5.** The results obtained by our RGB-CoGes method when using VideoMAE and VideoMAEv2 in isolated hand gesture recognition model, with and without post-processing, are presented.

The best values for each model are highlighted in bold.			
Model	Post-processing	Frame-wise Accuracy	Score
VideoMAE	X	0.772	0.423
	✓	0.870	0.681
VideoMAEv2	X	0.871	0.677
	✓	0.872	0.678

because, during the break time called ‘No Gesture’, subjects may perform any action they want.

3) CONTINUOUS GESTURE RECOGNITION RESULTS

a: RESULTS ON CoboGesture DATASET

Table 6 illustrates a comparative evaluation of the proposed methods RGB-CoGes and SkeRGB-CoGes against state-of-the-art models on the CoboGesture. On the CoboGesture dataset, the proposed method RGB-CoGes significantly surpasses skeleton-based counterparts. It achieves a frame-wise accuracy of 87% and an activity overlap score of 0.68, outperforming the continuous method based on DD-Net,

TDDNet, ST-GCN and CTR-GCN, which attain lower scores of 60% and 0.55, 66% and 0.59, 59% and 0.41, 74% and 0.57 for frame-wise accuracy and activity overlap score respectively. Although skeleton-based models are lightweight and fast, with CTR-GCN achieving competitive results, the RGB-based method still has stronger recognition capability. To take advantage of the lightweight nature of skeleton, we combined skeleton and RGB data in SkeRGB-CoGes, which achieved a frame-wise accuracy of 90% and an activity overlap score of 67%. These results are comparable to the RGB-based method, indicating that the framework performs reliably and is ready for online experiments.

b: RESULTS ON IPN-HAND AND EgoGesture DATASETS

For the IPN-Hand dataset, the proposed method also shows strong performance with 80% accuracy and a 0.37 overlap score. The performance of the skeleton-based DD-Net model is lower, with only 57% accuracy. From Table 7, we can see that our method achieves a frame-wise accuracy of 0.56 and a score of 0.36 on the EgoGesture dataset. These results are lower than the results on other datasets. The reason is that the gestures in EgoGesture are short. The mean

TABLE 6. The comparison between the proposed methods (RGB-CoGes and SkeRGB-CoGes) and the state-of-the-art methods on CoboGesture dataset. The best values are highlighted in bold.

Dataset	Models	Modality	Frame-wise Accuracy	Score
CoboGesture	Continuous method based on DD-Net [49]	Skeleton	0.60	0.55
	Continuous method based on TDDNet [46]	Skeleton + Text	0.66	0.59
	Continuous method based on ST-GCN [47]	Skeleton	0.59	0.41
	Continuous method based on CTR-GCN [48]	Skeleton	0.74	0.57
	RGB-CoGes (our)	RGB	0.87	0.68
	SkeRGB-CoGe (our)	RGB + Skeleton	0.90	0.67

TABLE 7. The comparison between the proposed methods (RGB-CoGes and SkeRGB-CoGes) and the state-of-the-art methods on IPN-Hand and EgoGesture datasets. The best values are highlighted in bold.

Dataset	Models	Modality	Frame-wise Accuracy	Score
IPN-Hand [30]	Continuous method based on DD-Net [49]	Skeleton	0.57	-
	RGB-CoGes with VideoMAE (our)	RGB	0.80	0.37
	RGB-CoGes with VideoMAEv2 (our)	RGB	0.83	0.51
EgoGesture [38]	RGB-CoGes with VideoMAEv2 (our)	RGB	0.56	0.36

TABLE 8. Online evaluation results on edge device.

Evaluation metrics	Values
Frame-wise Accuracy	0.94
Score (activity overlap score)	0.56

duration of gestures is about 38 frames [38]. Short gestures give little temporal context for models that rely on motion patterns. With short clips, a sliding-window classifier may see only part of a gesture. This makes it harder to learn key temporal features and to classify the correct label. Second, the EgoGesture dataset is recorded from a first-person viewpoint. In this setting, the camera moves together with the performer, which introduces strong background motion and blur. The hands often appear only partially in the frame or become occluded, and the narrow field of view causes unusual perspectives. These factors make it difficult for the model to extract consistent spatial features. Moreover, this viewpoint is different from the typical third-person view used in human-robot interaction, making the dataset less suitable for evaluating models intended for cobot applications. Overall, the results highlight that the robustness of the proposed method.

4) DEPLOYMENT ON EDGE DEVICE AND ONLINE EVALUATION

For online deployment, we move the framework to an edge device and conduct the experiment on three new subjects, in a robotics lab, under conditions closely resembling an industrial setting as shown in Figure 12a. Each subject performs 19 gestures in the COBOT dataset continuously, with a short break between each gesture. The deployed edge device is a Jetson AGX Orin Developer Kit version 32GB, equipped with JetPack 6.2, CUDA 12.6 and running Ubuntu 22.04. For capturing and processing actions in real time, an Intel RealSense camera operated at a resolution of 640×480 and a frame rate of 30 FPS is connected to the



(a) Robotics lab environment



(b) Setup of the online experiment

FIGURE 12. Deployment of the framework in industrial setting.

Jetson as presented in Figure 12b. In terms of framework configuration, Mediapipe is not yet officially supported on Jetson devices, so the pose estimation model runs on the CPU. The VideoMAE model, on the other hand, takes pre-trained weights from isolated recognition, and is deployed on CUDA to leverage GPU acceleration. The window size is set to 32 frames and the step is 10 frames during the prediction. The results of the experiment are presented in Table 8.

The frame-wise accuracy is evaluated on the processed frames, while the activity overlap score is computed using the actual timestamps of these frames. The framework achieves high accuracy per frame of 94%, demonstrates a strong prediction capability of VideoMAE as previously demonstrated. However, the overlap score of 56% is quite modest. During the experiment, we calculate the processing speed of both Mediapipe and VideoMAE, the results are approximately 4.5 FPS and 2.5 FPS, respectively. Compared to the capturing rate of the camera, the low processing speed indicates a bottleneck that may result in dropped frames. In other words, two consecutive frames recorded may be processed several seconds apart in real time, which can explain the gap between two metrics. For further investigation, we compute the average recognition latency of one subject, measured from the end of the performed action to the framework's output. The result is approximately 1.3s, with occasional spikes in latency. These results indicate that using skeleton-based boundary detection and simple voting mechanism enables real-time response; however, delays may occur if a subject completes an action while VideoMAE is still processing the current input frames. These occasional latency spikes often exceed 2s, as a result, will be penalized in the activity overlap score. Nevertheless, this setup still allows for the evaluation of the model's capability and serves as a basis for future improvements.

VI. CONCLUSION AND FUTURE WORKS

In this study, we presented a continuous gesture recognition dataset as well as a method for recognizing these hand gestures in human-collaborative robot interaction. The proposed method integrates VideoMAE, sliding window, and post-processing techniques. Due to the lack of available datasets for continuous gesture recognition in the context of human-collaborative interaction, we collected and fully annotated a new dataset named CoboGesture. This dataset will be made available to the research community to promote future research in human-collaborative interaction. Extensive experiments were conducted to assess the performance of the proposed method in both isolated and continuous recognition tasks. The results confirm the robustness of the proposed method, achieving 95.4% of accuracy for isolated recognition, 87% of Frame-wise Accuracy, and 0.68 of activity overlap score for continuous recognition in CoboGesture. On the IPN-Hand dataset, the proposed method achieves an accuracy of 78.81% for isolated gesture recognition. For the continuous gesture recognition task, it also outperforms the DDNet-based approach, achieving a frame-wise accuracy of 80% and a score of 0.37. Despite these promising results, the proposed method still demands considerable computational resources. Therefore, model optimization techniques such as model quantization and model pruning should be investigated to enable real-time performance, especially given the limited computing capabilities typically found in robotic systems. Second, end-to-end recognition models could be applied to simultaneously detect the start and end timestamps

and classify gestures. Finally, since human communication involves multiple modalities, human-robot interaction should integrate other modalities, such as speech, in addition to gestures.

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